Esercizio 1)

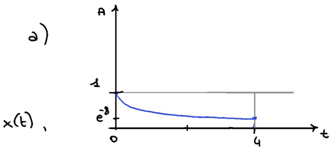
Dato il segnale continuo nel tempo x(t)= Π $\left(\frac{t-2}{4}\right) e^{-2t}$

a) Determinarne la rappresentazione grafica

b) Verificare se x(t) è un segnale di energia o di potenza e calcolare la corrispondente grandezza

c) Determinare l'espressione analitica e grafica dei seguenti segnali derivati da x(t): y(t) = - x (-t), z(t)= x(t+4)

a)



$$x(t) = \overset{A}{1}\,\Pi\left(\frac{\overset{\text{ritardo}}{t-2}}{\underset{\text{larghezza}}{4}}\right)\boxed{e^{-2t}} \to \text{decrescente}$$

$$x(t) = \begin{cases} e^{-2t} & 0 \le t \le 4 \\ 0 & \text{, altrove} \end{cases} \qquad \begin{matrix} x(0) = 1 \\ x(4) = e^{-8} \end{matrix}$$

b) È un segnale di energia perché il suo $\int$ è un valore finito

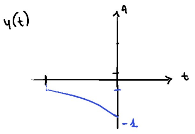
$$E_x = \int_{\mathbb{R}} x^2(t)\,dt = \int_0^4 (e^{-2t})^2\,dt = \int_0^4 e^{-4t}\,dt = \int f'(x)\cdot e^{f(x)} = e^{f(x)}$$

$$= -\frac{1}{4}\int_0^4 \underset{f'(x)}{-4}\, e^{\overset{\to f(x)}{-4t}} = -\frac{1}{4}\left[e^{-4t} - e^{-4t}\right]_0^4$$

$$= -\frac{1}{4}\left(e^{-16} - 1\right) = 0{,}25\text{ J}$$

c) $y(t) = -x(-t)$ $\qquad z(t) = x(t+4)$

$$y(t) = -\Pi\left(\frac{-t-2}{4}\right)e^{-2(-t)} = -\Pi\left(-\frac{(t+2)}{4}\right)e^{2t}$$



$$y(t) = \begin{cases} -e^{2t} & , -4 \le t \le 0 \\ 0 & , \text{altrove} \end{cases}$$

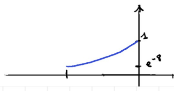
$y(0) = -1 \qquad y(-4) = -e^{-8}$

$$E_x = \int_{\mathbb{R}} x^2(t)\,dt = \int_{-4}^{0} e^{4t}\,dt = \frac{1}{4}\int_{-4}^{0} 4e^{4t}\,dt = \frac{1}{4}\left[e^{4t} - e^{4t}\right]_{-4}^{0}$$

$$= 0{,}25\text{ J}$$

$z(t) = x(t+4)$

$$z(t) = \Pi\left(\frac{t+4-2}{4}\right)e^{-2(t+4)} = \Pi\left(\frac{t+2}{4}\right)e^{-2t-8}$$

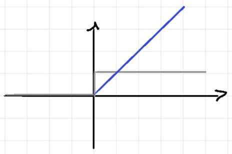


$$z(t) = \begin{cases} e^{-2t-8} & , -4 \le t \le 0 \\ 0 & , \text{altrove} \end{cases}$$

Esercizio 4

Studiare le proprietà di simmetria del segnale x(t) = t * u(t)

$x(t) = t \cdot u(t)$ $\qquad u(t) = \begin{cases} 1, & t \ge 0 \\ 0, & \text{altrove} \end{cases}$



→ No simmetrico

Propietà simmetria

Pari:

- $x(-t) = x(t)$

  $-t \cdot u(-t) = t \cdot u(t)$

  $-u(-t) = u(t)$

  ⇓

  Falso

Dispari

- $x(-t) = -x(t)$

  $-t \cdot u(-t) = t(-u(t))$

  $-tu(-t) = -tu(-t)$

  ⇓

  Falso

Non ha proprietà simmetriche.

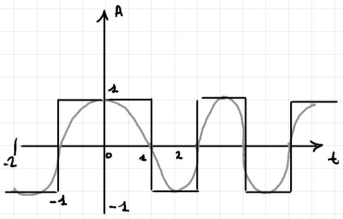
Esercizio 2)

2) Si consideri il segnale: $s(t) = \text{sgn}\left(A \cdot \cos\frac{2\pi t}{T_0}\right)$

a) Rappresentare graficamente il segnale ponendo A=0.5 V e T0=4 sec

b) Determinare se s(t) è un segnale di energia o di potenza

c) Calcolare l'energia del segnale in un periodo e la potenza media



$$\text{sgn}(x) = \begin{cases} 1, & x \ge 0 \\ -1, & x < 0 \end{cases}$$

a) $s(t) = \text{sgn}\left(\frac{1}{2}\cos\left(\frac{2\pi t}{4s}\right)\right)$ $\qquad s(-1) =$

$s(t) = 1$ quando $\cos\frac{2\pi t}{T_0} \ge 0$: $\qquad \frac{1}{2}\cos\left(\frac{\pi}{2}t\right) \to 0$

$s(1) = 0 \quad s(2) = -1$

$\frac{3}{2}\pi$ / $\quad -\frac{\pi}{2} + k2\pi \le \frac{2\pi t}{T_0} \le \frac{\pi}{2} + k2\pi$

$A = 1/2\ , T_0 = 4$

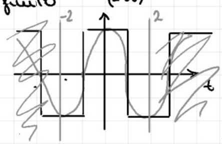
$$2\left(-\frac{\pi}{2} + 2k\pi\right) \le \frac{\cancel{2}\pi t}{\cancel{4}2} \le \left(\frac{\pi}{2} + k2\pi\right)\cdot 2$$

$\pi \quad \frac{\pi}{2} \quad 2\pi \quad \frac{3}{2}\pi$

$$\frac{\cancel{\pi}(4k-1)}{\cancel{\pi}} \le \cancel{\pi}t \le \frac{\cancel{\pi}(1+4k)}{\cancel{\pi}}$$

$$\underbrace{4k-1 \le t \le 1+4k}_{\text{dove } s(t) \text{ è positivo}} \qquad k \in \mathbb{Z}$$

b) È un segnale di potenza poiché l'integrale non è finito (=∞)



$$P_s = \lim_{T\to\infty}\frac{1}{T_0}\int_{-T/2}^{T/2} s^2(t)\,dt = \quad s(t)\begin{cases} 1, & \text{se } t \ge 0 \\ -1, & \text{se } t \le 0 \end{cases}$$

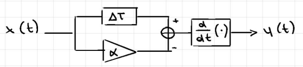
1 / -1

$$P_s = \lim_{T\to\infty}\frac{1}{4}\int_{-2}^{2} s^2(t)\,dt = \frac{1}{4}\cdot \overset{2-(-2)}{4} = 1\,\text{W}$$

$$E_{sp} = \int_{-2}^{2} s^2(t)\,dt = 4\text{ J}$$

Esercizio 1

Determinare se il sistema descritto dal seguente schema a blocchi è LTI



Il sistema è una combinazione lineare di funzioni LTI (ritardo, guadagno, derivazione), quindi è un sistema LTI

Esercizio 2

Determinare se il sistema che produce la seguente elaborazione del segnale x(t) è LTI

$$y(t) = 2x(t)\cos(t)$$

y(t) = 2x(t)cos (t)

- Lineare:

$f(x_1(t)) = y_1(t) = 2x_1(t)\cos(t)$

$f(x_2(t)) = y_2(t) = 2x_2(t)\cos(t)$

$ax_1(t) + bx_2(t) = 2[ax_1(t) + bx_2(t)]\cos$

$\overset{!}{=} ay_1(t) + by_2(t) \Rightarrow$ è lineare

- Tempo invarianza:

$f(x(t)) = y(t) \Rightarrow f(x(t-t_0)) = y(t-t_0)$

$x(t-t_0) = 2x(t-t_0)\cos(t) \ne y(t-t_0) = 2x(t-t_0)\cos(t-t_0)$

⇒ Non è tempo invariante ⇒ Non è LTI

Esercizio 3

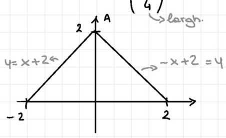
Dato il segnale x(t) = 2Λ $\left(\frac{t}{4}\right)$

a) si fornisca la rappresentazione grafica - se ne calcoli energia e potenza media b)

c) ~~b)~~ si rappresenti graficamente il segnale v(t)=x(2t)

$$x(t) = \overset{A}{2}\,\Lambda\left(\frac{t}{\underset{\text{largh.}}{4}}\right) \quad \text{centrato in } 0$$

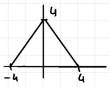
$$x(t) = \begin{cases} \frac{2(2-t)}{2} & , 0 \le t \le 2 \\ (2+t) & , -T/2 \le t < 0 \\ 0 & , \text{altrove} \end{cases}$$



b) Si tratta di energia del segnale, perché l'integrale è un valore finito

$$E_x = \int_{\mathbb{R}} x^2(t)\,dt$$

$$= 2\int_0^2 (2-t)^2\,dt$$



$$= 2\int_0^2 4 - 4t + t^2\,dt$$

$$= 2\left[\int_0^2 4\,dt - \int_0^2 4t\,dt + \int_0^2 t^2\,dt\right]$$

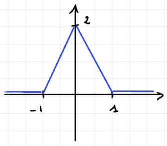
$$= 2\left\{4(2-0) - 4\left[\frac{t^2}{2}\right]_0^2 + \left[\frac{t^3}{3}\right]_0^2\right\}$$

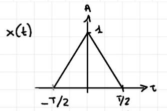
$$= 2\left(8 - 8 + \frac{8}{3}\right) = \frac{16}{3} \simeq 5{,}3\text{ J}$$

c) $v(t) = x(2t)$ $\qquad x(t) = 2\Lambda(t/4)$

$$v(t) = 2\Lambda\left(\frac{\cancel{2}t}{\cancel{4}2}\right) = 2\Lambda\left(\frac{t}{2}\right)$$

$$v(t) = \begin{cases} 2(1-t) & , 0 \le t \le 1 \\ 2(1+t) & , -1 \le t \le 0 \\ 0 & , \text{altrove} \end{cases}$$





Esercizio 7

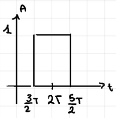
Siano dati i segnali x(t) = Λ $\left(\frac{t}{T}\right)$

e y(t) = Π $\left(\frac{t-2T}{T}\right)$

a) calcolarne la cross-correlazione

b) valutare in quale istante temporale assume valore massimo.

c) Che significato ha questo istante?



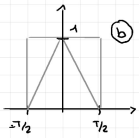
a) $R_{xy}(\tau) = \int_{\mathbb{R}} x(t)\,y(t+\tau)\,dt$

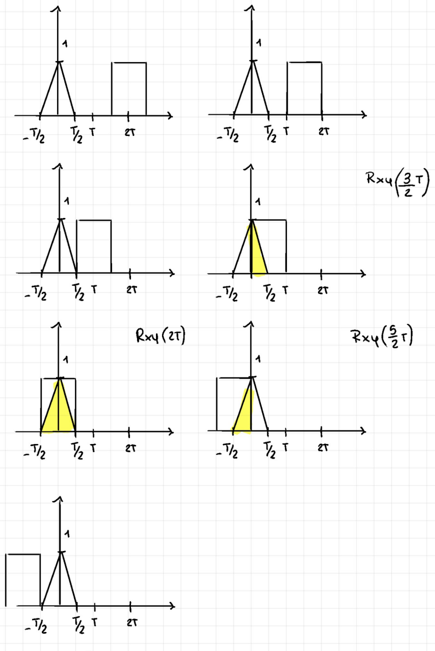
$x(t) = 1 - t/T/2 = 1 + \frac{2t}{T}$

$y(t) = 1 \quad$ per $|t| \le T/2$

$$\Pi\left(\frac{t+\tau-2T}{T}\right) = \begin{cases} A, \\ 0, & \text{altrove} \end{cases} \to |t+\tau-2T| < \frac{T}{2}$$

$\int_{\mathbb{R}} 1 + \frac{2t}{T} \cdot A$




1
-T/2
T/2
T
2T
$R_{xy}\left(\frac{3}{2}T\right)$
$R_{xy}(2T)$
$R_{xy}\left(\frac{5}{2}T\right)$

$$R_{xy}(\tau) = \int_{\mathbb{R}} x(t)\,y(t+\tau)\,dt$$

CONSIDERANDO AMPIEZZA DEL Δ = 2

Esercizio 5

Dato il segnale $x(t) = 2\Pi\left(\frac{t}{T}\right)\Lambda\left(\frac{t}{2T}\right)$

a) si chiede di rappresentarlo graficamente,

b) calcolarne energia totale e potenza media,

c) verificando se sia un segnale di energia o di potenza.

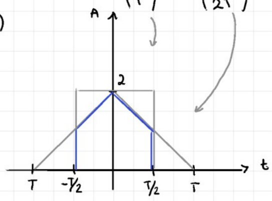
d) Si chiede inoltre di proporre una rappresentazione alternativa del segnale, basata sulla somma e non sul prodotto di segnali

$\underline{x(t)} = 2\Pi\left(\frac{t}{T}\right) 1\Lambda\left(\frac{t}{2T}\right)$

$\Delta = \frac{-2t}{T}+2$

$\Pi = 2t$

a)



$x(t) = \frac{-2t}{T}+2$
per $t \le T/2$

b) $E_x = \int_{\mathbb{R}} x^2(t)\,dt$

$= 2\int_0^{T/2}\left(\frac{-2t}{T}+2\right)^2 = 2\int_0^{T/2}\frac{4t^2}{T^2}-\frac{8t}{T}+4\,dt$

$= 2\cdot 4\left[\int_0^{T/2}\frac{t^2}{T^2}\,dt - \int_0^{T/2}\frac{2t}{T}\,dt + \int_0^{T/2}1\,dt\right]$

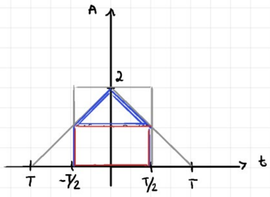
$= 8\left\{\frac{1}{T^2}\left[\frac{t^3}{3}\right]_0^{T/2} - \frac{2}{T}\left[\frac{t^2}{2}\right]_0^{T/2} + \left[T/2-0\right]\right\}$

$= 8\left(\frac{1}{\cancel{T^2}}\cdot\frac{T^{\cancel{3}}}{8\cdot 3} - \frac{\cancel{2}}{\cancel{T}}\cdot\frac{T^{\cancel{2}}}{4\cdot\cancel{2}} + \frac{T}{2}\right) = \cancel{8}\left(\frac{T-6T+12T}{\cancel{24}\,3}\right)$

$= \frac{7}{3}T \rightarrow \frac{T}{3}J$

d) Rappresentazione basata su somma:

$\underline{x(t)} = \Pi\left(\frac{t}{T}\right)+1\Lambda\left(\frac{t}{T}\right)$



$A = 1\Pi\left(\frac{t}{T}\right)$ $\quad x_A(t) = \begin{cases}1 & |t| \le T/2\\ 0 & \text{, altrove}\end{cases}$

$x_A(t) = \begin{cases}1t & ,t\le T/2\\ 0 & \text{, altrove}\end{cases}$

$E_{xA} = 2\int_0^{T/2} 1\,dt = T$

$B = \Lambda\left(\frac{t}{T}\right)$ $\quad x_B(t) = \begin{cases}1-\frac{t2}{T} & ,0\le t\le T\\ 0 & \text{, altrove}\end{cases}$
↪ solo un lato

$E_{xB} = 2\int_0^{T/2}\left(1-\frac{t2}{T}\right)^2 dt$

$= 2\int_0^{T/2} 1-\frac{4t}{T}+\frac{4t^2}{T^2}\,dt$

$= 2\left[\int_0^{T/2}1\,dt - \frac{4}{T}\int_0^{T/2}t + \frac{4}{T^2}dt\int_0^{T/2}t^2\,dt\right]$

$= 2\left\{\frac{T}{2} - \frac{\cancel{4}}{\cancel{T}}\cdot\frac{T^{\cancel{2}}}{42} + \frac{\cancel{4}}{\cancel{T^2}}\cdot\frac{T^3}{\underset{2}{\cancel{8}}\cdot 3}\right\} = 2\left(\frac{3T-3T+T}{6}\right)$

$= \frac{T}{3}$

$E_x = E_{xA}+E_{xB}+\underline{2_{xAB}} \rightarrow 2B$

$= T+\frac{T}{3}+2\left(\frac{T}{3}\right) = \frac{3T+3T}{3} = 2T$

↪ BOH
NON TORNA

$\int(f)^2 \qquad \int(\Pi+\Lambda)^2 \qquad \Pi^2+\Lambda^2+2\Pi\Lambda$

Esercizio 3

Determinare se il sistema che produce la seguente elaborazione del segnale x(t) è LTI $\quad y(t) = x(t)e^{x(t)}$

- Linearità

$x_1(t) = y_1(t) = x_1(t)e^{x_1(t)}$

$x_2(t) = y_2(t) = x_2(t)e^{x_2(t)}$

$a x_1(t)+b x_2(t) = a x_1(t)e^{a x_1(t)} + b x_2(t)e^{b x_2(t)}$

$a y_1(t)+b y_2(t) = a x_1(t)e^{x_1(t)} + b x_2(t)e^{x_2(t)}$ $\Big) \neq$

⇓
Non è lineare
Non è LTI

- Tempo invarianza:

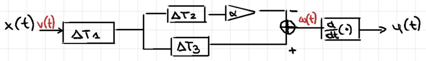
$x(t) = y(t) \Rightarrow x(t-t_0) = y(t-t_0)$

$x(t-t_0)e^{x(t-t_0)} = y(t-t_0) = x(t-t_0)e^{x(t-t_0)}$

è tempo invariante

Esercizio 5

Dato il sistema seguente, determinarne l'uscita in funzione dell'ingresso



$v(t) = x(t-T_1)$

$w(t) = v(t-T_3) - \alpha v(t-T_3)$

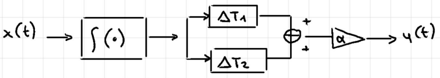
$\frac{d}{dt}w(t) = y(t)$

$y(t) = \frac{d}{dt}\left(v(t-T_3) - \alpha v(t-T_3)\right)$

$= \frac{d}{dt}\left(x(t-T_1-T_3) - \alpha x(t-T_1-T_3)\right)$

Esercizio 5

Calcolare la risposta impulsiva del sistema descritto dal seguente schema a blocchi



$y(t) = \int[x(t-\Delta T_1)+x(t-\Delta T_2)]\alpha\,dt$

$= \alpha\left[\int x(t-\Delta t_1)\,dt + \int x(t-\Delta t_2)\,dt\right]$

y(t) = uscita per un generico x(t). Se sostituisco δ(t) a x(t) ottengo h(t)

Esercizio 6

a) Determinare la risposta impulsiva h(t) del sistema che produce la seguente elaborazione del segnale x(t), e proporre uno schema a blocchi che implementi h(t) ↪ b)

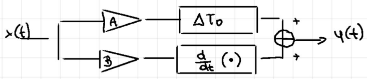
$$y(t) = Ax(t-T_0) + B\frac{d}{dt}x(t)$$

a) sostituisco $x(t)$ con $\delta(t)$

$h(t) = \Delta\delta(t-T_0) + B\frac{d}{dt}\delta(t)$ → doppietto $= \overset{\text{Area}}{\pm B}$

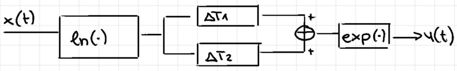
$= A\delta(t-T_0) + B\delta'(t)$

b)



Esercizio 4

Determinare se il sistema descritto dal seguente schema a blocchi è LTI



Questo sistema non è LTi

Exp e ln sono due operazioni non lineari

Operazioni non LTi:
- $t\cdot x(t)$ • $\sin(t)\cdot x(t)$ • $x(t)+t$

se ci fosse stato un solo blocco di ritardo sarebbe stato LTi.

$v(t) = \log(x(t))$

$w(t) = v(t-T_1)+v(t-T_2)$

$y(t) = e^{w(t)}$

$y(t) = e^{v(t-T_1)+v(t-T_2)}$

$= e^{\log x(t-T_1)+\log x(t-T_2)} = e^{\log x(t-T_1)}\,e^{\log x(t-T_2)}$

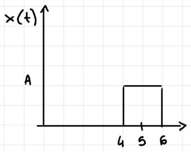
$= x(t-T_1)\,x(t-T_2)$

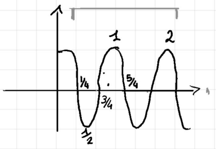
Trovo pti:
$2\pi t = \pi \Rightarrow \frac{1}{4}$

Esercizio 3

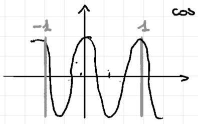
Calcolare la convoluzione tra i segnali x(t) e y(t) sotto definiti:

$x(t) = A\Pi\left(\frac{t-5}{2}\right) \qquad y(t) = \cos 2\pi t$

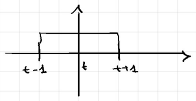




$y(t) = \int_{\mathbb{R}} y(\tau) * x(t-\tau) * A\delta(t-5)\,d\tau$



$\cos 2\pi(\tau)$



$\int_{t-1}^{t+1}\cos 2\pi(\tau)\,d\tau =$

$= \frac{1}{2\pi}\left[\sin(2\pi(t+1)) - \sin(2\pi(t-1))\right]$

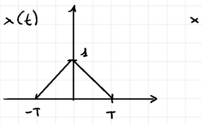
$= \frac{1}{2\pi}\left[\sin(2\pi t+2\pi) - \sin(2\pi t+2\pi)\right]$

$= \frac{1}{2\pi}\left[\sin(2\pi t)\overset{1}{\cos(2\pi)} + \cos(2\pi t)\overset{\rightarrow 0}{\sin(2\pi)} - \sin(2\pi t)\overset{1}{\cos(2\pi)} - \cos(2\pi t)\overset{\rightarrow 0}{\sin(2\pi)}\right]$

0

Esercizio 7

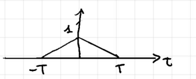
Un sistema LTI ha risposta impulsiva $h(t) = \delta(t) + 2\delta(t-T)$. In ingresso al sistema viene posto un segnale causale $x(t) = \Lambda(t/2T)$. Calcolare il segnale y(t) in uscita dal sistema.



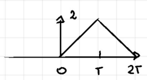
$x(t) = \begin{cases}1-\frac{1t}{2T} & ,0\le t\le 2T\\ 1+\frac{1t}{2T} & ,-2T\le t\le 0\end{cases}$

$y(t) = x(t)*h(t) = \int_{\mathbb{R}} x(\tau)\cdot\left[\delta(t-\tau)+2\delta(t-\tau-T)\right]d\tau$

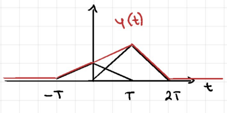
$x(\tau)\cdot\delta(t-\tau) = x(t) \qquad x(t)\cdot 2\delta(t-T-\tau)\,d\tau = x(t-T$



+



$y(t) = \Lambda\left(\frac{t}{2T}\right) + x(t-T) = \Lambda\left(\frac{t-T}{2T}\right)$

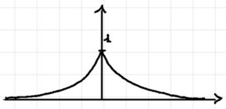


Esercizio 1

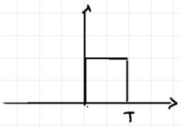
1) Calcolare la convoluzione tra i segnali x(t) e y(t) sotto definiti:

$x(t) = e^{-\beta|t|}$, $\beta > 0$ $y(t) = \Pi\left(\frac{t - T/2}{T}\right)$

$$x(t) = \begin{cases} e^{-\beta t} & , t > 0 \\ 1 & , t = 0 \\ e^{+\beta t} & , t < 0 \end{cases}$$



$$y(t) = \begin{cases} 1 & , t \leq T \\ 1 & , t > 0 \end{cases}$$

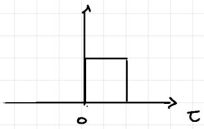


$y(t) = \int_{\mathbb{R}} x(\tau)\, h(t-\tau)\, d\tau$

Convoluzione grafica:

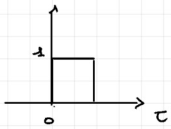
In τ

$$y(\tau) = \begin{cases} 1 & , \tau \leq t \\ 1 & , \tau > 0 \end{cases}$$

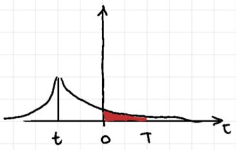


$$x(t-\tau) = \begin{cases} e^{-\beta(t-\tau)} & , \tau > 0 \\ 1 & , t = 0 \\ e^{-\beta(t-\tau)} & , t < 0 \end{cases}$$

per $t < 0$:



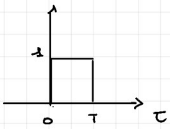
$$\int_0^T e^{-\beta(t-\tau)}\, d\tau = \int_0^T e^{-\beta t} \cdot e^{\beta\tau}\, d\tau$$



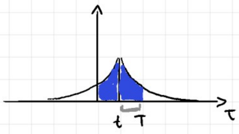
$$= e^{-\beta t} \int_0^T e^{\beta\tau}\, d\tau = e^{-\beta t}\left(-\frac{1}{\beta}\right)\left[e^{\beta T} - 1\right]$$

$$= e^{-\beta t} \cdot \frac{1}{\beta}\left[1 - e^{\beta T}\right]$$

per $0 < t < T$:



$$\int_0^T e^{-\beta(t-\tau)}\, d\tau$$



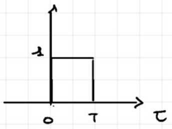
$$= \int_t^T e^{-\beta(t-\tau)}\, d\tau$$

$$= \int_t^T e^{-\beta t}\, e^{-\beta\tau}\, d\tau = e^{-\beta t}\int_t^{\tau} e^{-\beta\tau}\, d\tau = e^{-\beta t}\left(-\frac{1}{\beta}\right)\left[e^{-T^2} - e^{tT}\right]$$

$$= \frac{e^{-\beta t}}{\beta}\left[e^{tT} - e^{T^2}\right] = \frac{e^{-\beta t^2 T}}{\beta} - \frac{e^{-\beta t T^2}}{\beta} = 2\frac{e^{-\beta t T}}{\beta}\left(e^{t} - e^{T}\right)$$

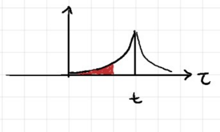
perché avevo fatto da t a T

per $t > T$:



$$\int_0^T e^{+\beta(t-\tau)}\, d\tau = e^{\beta t}\int_0^T e^{-\beta\tau}\, d\tau$$

$$= e^{\beta t} \cdot \frac{1}{\beta}\left[e^{-\beta T} - 1\right]$$

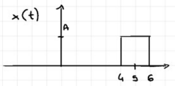


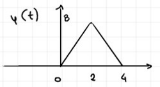
$$y(t) = \int_{\mathbb{R}} x(\tau)\, h(t-\tau)\, d\tau = \begin{cases} e^{-\beta t}\frac{1}{\beta}\left(1 - e^{\beta T}\right) & , t < 0 \\ 2\frac{e^{-\beta t T}}{\beta}\left(e^{t} - e^{T}\right) & , 0 \leq t \leq T \\ e^{\beta t} \cdot \frac{1}{\beta}\left(e^{-\beta t} - 1\right) & , t > T \end{cases}$$

Esercizio 2

Calcolare la convoluzione tra i segnali x(t) e y(t) sotto definiti:

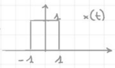
$x(t) = A\,\Pi\left(\frac{t-5}{2}\right)$ $y(t) = B\,\Lambda\left(\frac{t-2}{4}\right)$



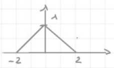


Convoluzione:

$x(t) = \Pi\left(\frac{t}{2}\right) * A\,\delta(t-5)$



$y(t) = \Lambda\left(\frac{t}{4}\right) * B\,\delta(t-2)$

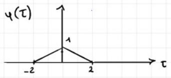


$y_R(t) = \int_{\mathbb{R}} x(\tau) \cdot y(t-\tau)\, d\tau = \Pi\left(\frac{t}{2}\right) * \Lambda\left(\frac{t}{4}\right) * AB\,\delta(t-7)$

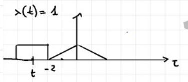
$$x(t) = \begin{cases} 1 & , t \geq -1 \\ 1 & , t \leq 1 \end{cases} \qquad y(t) = \begin{cases} 1 - \frac{1}{2}t & 0 < t \leq 2 \\ 1 + \frac{1}{2}t & , -2 \leq t \leq 0 \end{cases}$$

$\frac{2-t}{2}$ $\frac{1(2-|t|)}{2}$ $\frac{2+t}{2}$

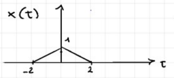
Tengo fermo il triangolo



x(t)= 1

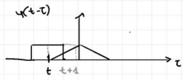


0, $t < -3$



per $-3 \leq t \leq -1$:

$$\int_{-2}^{t+1} 1 + \frac{1}{2}(\tau)\, d\tau =$$



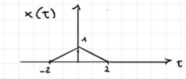
$$= t+1-(-2) + \frac{1}{2}\int_{-2}^{t+1}\tau\, d\tau$$

$$= t+3 + \frac{1}{2}\left[\frac{\tau^2}{2}\right]_{-2}^{t+1} =$$

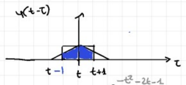
$$= t+3+\frac{1}{2}\left(\frac{(t+1)^2}{2} - \frac{(-2)^2}{2}\right) = t+3+\frac{1}{2}\left(\frac{t^2+2t+1}{2} - \frac{4}{2}\right)$$

$$= t+3+\frac{t^2}{4}+\frac{2t}{4}-\frac{3}{4} = \frac{t^2}{4}+\frac{6t}{4}+\frac{9}{4} = \frac{1}{4}\left(t^2+6t+9\right)$$

per $-1 \leq t \leq 1$:



$$\int_{t-1}^{0} 1 + \frac{1}{2}(\tau)\, d\tau =$$



$$= \left[-t+1 + \frac{1}{2}\int_0^1 \tau\, d\tau\right]$$

$$= \left[1-t + \frac{1}{2}\left[\frac{\tau^2}{2}\right]_{t-1}^{0}\right]$$

$-t^2-2t-1$

$$= 1-t+\frac{1}{2}\left(\frac{-(t-1)^2}{2}\right) = 1-t-\frac{t^2}{4}+\frac{2}{4}t-\frac{1}{4} = \boxed{-\frac{t^2}{4} - \frac{2t}{4} + \frac{3}{4}}$$

$$\int_0^{t+1} 1 - \frac{1}{2}(\tau)\, dt = t+1 \quad -\frac{1}{2}\int_0^{t+1}\tau\, d\tau =$$

$$= t+1-\frac{1}{2}\left[\frac{(t+1)^2}{2} - 0\right] = t+1-\frac{1}{4}t^2-\frac{2}{4}t-\frac{1}{4}$$

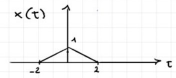
+

$$= -\frac{t^2}{4}+\frac{2}{4}t+\frac{3}{4} = \boxed{\frac{1}{4}\left(-t^2+2t+3\right)}$$

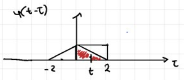
$$-\frac{t^2}{2} \qquad +\frac{6}{4} = \frac{1}{2}\left(-t^2+3\right)$$

=

per $1 \leq t \leq 3$



$$\int_{t-1}^{2} 1 - \frac{1}{2}\tau\, d\tau$$



$$= \left[2-(t-1)\right] - \frac{1}{2}\int \tau\, d\tau$$

$$= 3-t-\frac{1}{2}\left[\frac{\tau^2}{2}\right]_{t-1}^{2}$$

$$= 3-t-\frac{1}{2}\left[\frac{4}{2} - \frac{(t-1)^2}{2}\right] = 3-t-\frac{1}{2}\left[\frac{4}{2} - \frac{t^2-2t+1}{2}\right]$$

$$= 3-t-\frac{1}{2}\left[\frac{-t^2+2t+3}{2}\right] = 3-t+\frac{t^2}{4}-\frac{2t}{4}-\frac{3}{4} = \frac{1}{4}\left(t^2-6t+9\right)$$

$y_R(t) = \int_{\mathbb{R}} x(\tau) \cdot y(t-\tau)\, d\tau = \Pi\left(\frac{t}{2}\right) * \Lambda\left(\frac{t}{4}\right) * AB\,\delta(t-7)$

$$= \begin{cases} 0 & , t < -3 \\ \frac{1}{4}\left(t^2+6t+9\right) & , -3 \leq t \leq -1 \\ \frac{1}{2}\left(-t^2+3\right) & , -1 \leq t \leq 1 \\ \frac{1}{4}\left(t^2+6t+9\right) & , 1 \leq t \leq 3 \\ 0 & , t > 3 \end{cases}$$

vertice $\frac{3}{2}$ ∩

$$y_R(t) = \begin{cases} AB \cdot 0 & , (t-7) < -3 \Rightarrow t < 4 \\ AB\left[\frac{1}{4}\left[(t-7)^2+6(t-7)+9\right]\right] & , \begin{matrix} -3 \leq t-7 \leq -1 \\ 4 \leq t \leq 6 \end{matrix} \\ AB\left[\frac{1}{2}\left[-(t-7)^2+3\right]\right] & , \begin{matrix} -1 \leq t-7 \leq 1 \\ 6 \leq t \leq 8 \end{matrix} \\ AB\left[\frac{1}{4}\left[(t-7)^2+6(t-7)+9\right]\right] & , \begin{matrix} 1 < t-7 \leq 3 \\ 8 \leq t \leq 10 \end{matrix} \\ 0 & , \begin{matrix} t-7 > 3 \\ t > 10 \end{matrix} \end{cases}$$

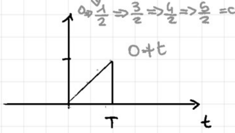
Esercizio 4

$$\begin{cases} \frac{t}{T} & , \begin{matrix} t \leq T/2 \\ t > -T/2 \end{matrix} \end{cases}$$

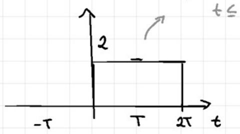
Calcolare la convoluzione z(t) tra i seguenti segnali x(t) e y(t), sotto definiti:

$x(t) = \frac{t}{T}\,\Pi\left(\frac{t - T/2}{T}\right)$ $y(t) = 2\,\Pi\left(\frac{t-T}{2T}\right)$

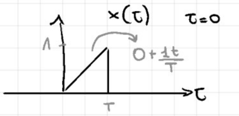
$0 \Rightarrow \frac{1}{2} \Rightarrow \frac{3}{2} \Rightarrow \frac{4}{2} \Rightarrow \frac{6}{2}$ = cresce

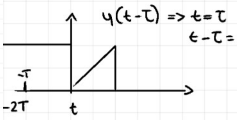


2, $t \geq 0$
$t \leq 2T$



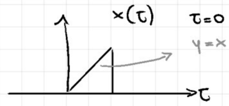
$y(t) = \int_{\mathbb{R}} x(\tau)\, h(t-\tau)\, d\tau = \frac{t}{T}\left(\frac{t}{T}\right) * \Pi\left(\frac{t}{2T}\right) * 2\delta(t-3T)$



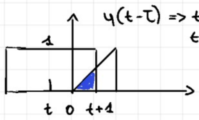


$y(t-\tau) \Rightarrow t = \tau$
$t - \tau = 2T \Rightarrow t - 2T = \tau$

$\Rightarrow$ 0, $t \leq -T$

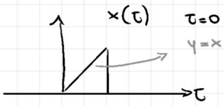


$-T \leq t \leq 0$

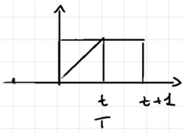


$y(t-\tau) \Rightarrow t = \tau$
$t - \tau = 2T \Rightarrow t - 2T = \tau$

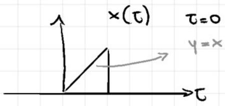
$$\frac{1}{T}\int_0^{t+1}\tau\, d\tau = \frac{(t+1)^2}{2} = \frac{t^2+2t+1}{2T}$$



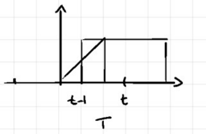
$0 \leq t \leq T$



$$\frac{1}{T}\int_0^T \tau\, d\tau = \frac{\tau^2}{2} = \frac{T^{\cancel{2}}}{2\cancel{T}} - 0$$



$T \leq t \leq 2T$



$$\int_{t-1}^{T}\tau\, d\tau = \frac{\tau^2}{2} = \frac{T^2}{2} - \frac{(t-1)^2}{2}$$

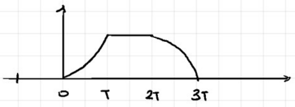
$$= \frac{T^2}{2} - \frac{t^2-2t+1}{2} = \frac{1}{2T}\left(T^2-t^2+2t-1\right)$$

0, $t > 2T$

$(t) = \int_{\mathbb{R}} x(\tau)\, h(t-\tau)\, d\tau = \frac{t}{T}\left(\frac{t}{T}\right) * \Pi\left(\frac{t}{2T}\right) * 2\delta(t-T)$

$$\begin{cases} 0, & t - T \leq -T \Rightarrow t \leq 0 \\ = \frac{2}{2T}\left[(t-T)^2 + 2(t-T) + 1\right] & , -T \leq t - T \leq 0 \Rightarrow 0 \leq t \leq T \\ \frac{2T}{\cancel{2}} & , T \leq t \leq 2T \\ \cancel{2}\left[\frac{1}{\cancel{2}T}\left(T^2 - t^2 + 2t - 1\right)\right] & , 2T \leq t \leq 3T \\ 0 & , t > 3T \end{cases}$$

t - T

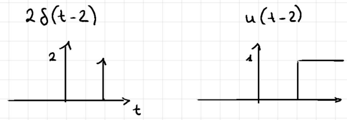


$h(t) = 0 \ , \ \forall t < 0$ = Non prevedibile

Esercizio 8

Dato il sistema LTI caratterizzato dalla risposta impulsiva h(t) sotto riportata, verificare che il sistema sia causale e calcolare la risposta ad un gradino unitario.

$h(t) = 2\delta(t-2) + u(t-2)$



per $t < 0 = 0 \Rightarrow$ causale

$$u(t) = \begin{cases} 1 & t \geq 0 \\ 0 & \text{, altrove} \end{cases}$$

$$y(t) = h(t) * u(t) = u(t) * [2\delta(t-2) + u(t-2)]$$

$u(\tau) \cdot h(t-\tau)\, d\tau$

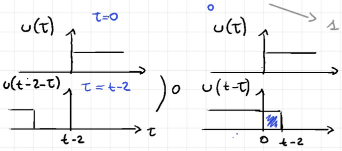
$$= \int_{\mathbb{R}} u(\tau) \cdot [2\delta(t-\tau-2) + u(t-\tau-2)]\, d\tau$$

$$= \overset{y_1}{\int_{\mathbb{R}} u(\tau) \cdot 2\delta(t-2-\tau)\, d\tau} + \overset{y_2}{\int_{\mathbb{R}} u(\tau) \cdot u(t-2-\tau)\, d\tau}$$

$$= \overset{y_1}{2u(t-2)} + \int_0^{t-2} u(\tau) \cdot u(t-2-\tau)\, d\tau$$

$\rightarrow 1 \ , t > 2$

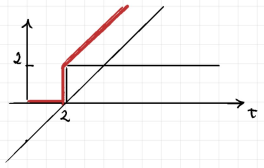
$\rightarrow 1 \ , t > 0$

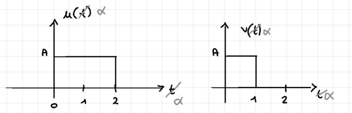


$0, \ t \leq 2$ $\quad$ $t \geq 2$

$$\int_0^{t-2} 1\, d\tau = \begin{cases} t-2 & , \ t \geq 2 \\ 0 & , \ t < 2 \end{cases}$$

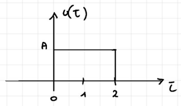
$$y(t) = 2u(t-2) + t - 2$$



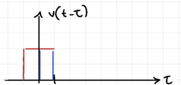


convolution: $y(t) = \int_{\mathbb{R}} u(\tau) v(t-\tau)\, d\tau$

1. Dominio $t \rightarrow \alpha$



2) trasformo in funzione di τ



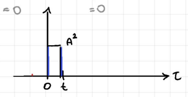
3) Prendi il valore t

$v(\alpha) = v(t-\tau)$

- $\alpha = 0 \ = \ t - \tau = 0 \Rightarrow t = \tau$

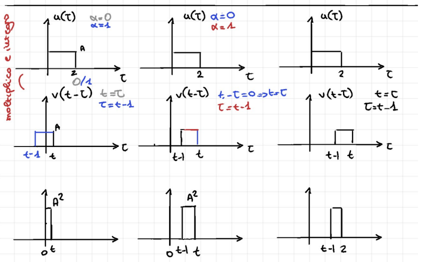
$\alpha = 1 \ = \ t - \tau = 1 \Rightarrow t - 1$

4) Moltiplica e integra



Area: $A^2 t$

5) Ripeti 2-4 ,



Area = $A^2 \cdot t$ $\quad$ Area = $A^2(t - t + 1) = A^2$ $\quad$ Area = $A^2(2 - t + 1) = A^2(3-t)$

0	$A^2 t$	A^2	$A^3(3-t)$	0
$t<0$	$t>0$ $t-1<0$ $0<t<1$	$t<2$ $t-1>0$ $1<t<2$	$t>2$ $t-1>2$ $2<t<3$	$t>3$

Esercizio 9

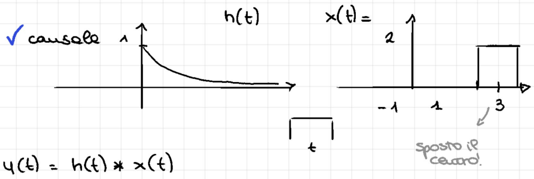
$h(t) = e^{-t} u(t)$

Un sistema LTI causale ha risposta all'impulso $h(t) = e \quad u(t)$.

Si vuole calcolare la risposta del sistema quando in ingresso sia posto un segnale rettangolare $x(t) = 2\Pi\left(\frac{t-3}{2}\right)$

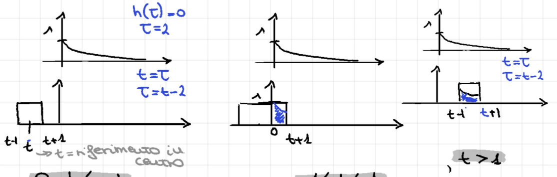
$x(t) = 2\Pi$

✓ causale



$$y(t) = h(t) * x(t)$$

$$y(t) = \int_{\mathbb{R}} h(\tau) \cdot x(t-\tau) \cdot 2\delta(t-3)\, d\tau$$



$0, \ t < -1$ $\quad$, $-1 \leq t \leq 1$ $\quad$, $t > 1$

$$\int_0^{t+1} e^{-\tau}\, d\tau = -1 \int_0^{t+1} e^{-\tau}\, d\tau$$

$$= -1\left(e^{-t-1} - 1\right) = 1 - e^{-(t+1)}$$

$$\int_{t-1}^{t+1} e^{-\tau}\, d\tau = -1\left[e^{-\tau}\right]_{t-1}^{t+1}$$

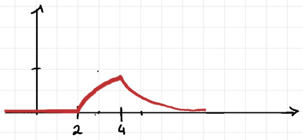
$$= -1\left(e^{-(t+1)} - e^{-(t-1)}\right)$$

$$= e^{-t+1} - e^{-t-1}$$

$$= e^{-t+3+1} - e^{-t+3-1}$$

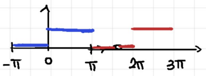
$$= e^{-t}(e^4 - e^2)$$

$$y(t) = \begin{cases} 0 \ , & t-3 < -1 \Rightarrow t < 2 \\ 2\left[1 - e^{-(t-2)}\right], & -1 \leq t-3 \leq 1 \Rightarrow 2 \leq t \leq 4 \\ 2\left[e^{-t}(e^4 - e^2)\right] \ , & t > 1 \Rightarrow t > 4 \end{cases}$$



SERIE DI FOURIER – ESERCIZIO

$$f(x) = \begin{cases} 0 & -\pi < x < 0 \\ 1 & 0 \leq x \leq \pi \end{cases}$$



Periodo : 2π $\quad$ è regolare a tratti $\Rightarrow$ ok serie

La funzione non è ne dispari, ne pari:

Trovo a_0 , a_n , b_n

$$s(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{2\pi n}{T} x + b_n \operatorname{sen} \frac{2\pi n}{T} x$$

$$a_0 = \frac{2}{2\pi} \int_{-\pi}^{\pi} f(x)\, dx = \frac{1}{\pi}\left[\int_{-\pi}^{0} 0\, dx + \int_0^{\pi} 1\, dx\right]$$

$$= \frac{1}{\pi}\, \pi = 1$$

$a_0 = \frac{2}{T} \int_{-T/2}^{T/2} f(x)\, dx$

$$a_n = \frac{2}{T} \int_{-T/2}^{T/2} f(x) \cos \frac{2\pi n}{T} x\, dx = \frac{1}{\pi}\left[\int_{-\pi}^{0} 0 + \int_0^{\pi} 1 \cos \frac{\cancel{2\pi} n}{\cancel{2\pi}} x\, dx\right]$$

$$= \frac{1}{n\pi}\left[\sin nx\right]_0^{\pi} = 0$$

$$b_n = \frac{2}{T} \int_{-T/2}^{T/2} f(x) \operatorname{sen} \frac{2\pi n}{T} x\, dx = \frac{1}{\pi}\left[\int_{-\pi}^{0} 0 + \int_0^{\pi} 1 \cdot \operatorname{sen} \frac{\cancel{2\pi} n x}{\cancel{2\pi}} \cdot dx\right]$$

$$= -\frac{1}{n\pi}\left[\cos nx\right]_0^{\pi} = -\frac{1}{n\pi}\left[(-1)^n - 1\right] \begin{cases} 0 \ , & n = \text{pari} \\ \frac{+2}{\pi n} \ , & n = \text{dispari} \end{cases}$$

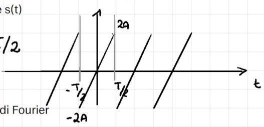
La serie trigonometrica è:

$$s(x) = \frac{1}{2} + \frac{2}{\pi}\left(\operatorname{sen} x + \frac{1}{3} \operatorname{sen} 3x + \frac{1}{5} \operatorname{sen} 5x + \frac{1}{2p+1} \operatorname{sen}(2p+1) \cdot x\right.$$

Esercizio 1

Sia data una funzione periodica x(t) di periodo T a 'dente di sega' in cui il periodo base è definito dalla funzione s(t)

$$s(t) = \frac{2A}{T} t, \quad |t| < T/2$$



a - Calcolare i coefficienti della serie di Fourier

b - Scrivere l'espressione analitica degli spettri di ampiezza e di fase

c - Rappresentarli graficamente

d - Calcolare la potenza media relativa all'armonica fondamentale del segnale x(t)

$$x(t) = s(t) * \sum_{\mathbb{R}} \delta(t - kT)$$

treno di impulsi

Periodo: T

$f_0 = 1/T$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{2\pi n t}{T}\right) + b_n \sin\left(\frac{2\pi n t}{T}\right)\right]$$

$a_0 = 0$ = valor medio

$$x(t) = \sum_{k=1}^{\infty} X_n e^{\frac{j2\pi nt}{T}}$$

$$X_n = \begin{cases} \frac{a_n}{2} e^{j\theta n} & n > 0 \\ \frac{a_{-n}}{2} e^{j\theta_{-n}} & n < 0 \\ a_n & n = 0 \end{cases}$$

$$X_n = \frac{1}{T} \int_{-T/2}^{T/2} x(t) e^{\frac{-i2\pi nt}{T}}\, dt$$

per $n = 0$:

$$X_0 = \frac{1}{T} \int_{-T/2}^{T/2} \frac{2At}{T} e^{\frac{-i2\pi 0 t}{T}}\, dt = \frac{2A}{T^2} \int_{-T/2}^{T/2} t\, e^{0}\, dt$$

$$= \frac{2A}{T^2}\left[\frac{t^2}{2}\right]_{-T/2}^{+T/2} = \frac{2A}{T^2}\left[\frac{\left(\frac{T^2}{4}\right)}{2} - \frac{\left(\frac{T^2}{4}\right)}{2}\right] = 0$$

$t = \frac{2T}{-i2\pi n}$

$y = \frac{-i2\pi nt}{T}$

$dy = \frac{-i2\pi n}{T} dt$

$$X_n = \frac{1}{T} \int_{-T/2}^{T/2} \frac{2A}{T} t\, e^{\frac{-i2\pi nt}{T}}\, dt =$$

$$= \frac{2A}{T^2}\left[t\left(\frac{1}{\frac{-i2\pi n}{T}} e^{\frac{-i2\pi nt}{T}}\right) - \int_{-T/2}^{T/2} \frac{t^2}{2}\left[\frac{1}{\frac{-i2\pi n}{T}} e^{\frac{-i2\pi nt}{T}}\right] dt\right.$$

$$= \frac{2A}{T^2} \int_{-T/2}^{\overset{-T/2}{T/2}} \frac{2T}{-\cancel{i2\pi n}} e^{y} \frac{-\cancel{i2\pi n}}{T}\, dy$$

$$= \frac{2A}{T^2} \cdot \frac{2\cancel{T}}{\cancel{T}} \int_{-T/2}^{T/2} e^{y}\, dy = \frac{4A}{T^2}\left[e^{T/2} - e^{-T/2}\right]$$

??

$$\ldots = \frac{\cancel{2}A}{\cancel{T^2}}\left\{-\frac{\cancel{T^2}}{\cancel{2}jn\pi} \cos n\pi - \frac{2j}{(2\pi n/T)} \sin n\pi\right\}$$

sempre nullo

$$X_n = -\frac{A}{jn\pi} \cos n\pi \begin{cases} \text{per } n \text{ pari} = -\frac{A}{jn\pi} \\ \text{per } n \text{ dispari} = \frac{A}{jn\pi} \end{cases}$$

$$\text{b)} \quad \begin{cases} \|X_n\| = \frac{A}{\pi |n|} \\ \arg X_n = \operatorname{sign}\left(\frac{(-1)^n}{n}\right) \frac{\pi}{2} \end{cases}$$

FUNZIONE PERIODICA → SERIE DI FOURIER

periodo $T \neq 0$

periodica = $f(x) = f(x + kT)$

$$s(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{2\pi n}{T} x + b_n \operatorname{sen} \frac{2\pi n}{T} x$$

nell'intervallo $[-T/2, T/2]$ $\quad$ $f(x) \sim s(x)$

$$a_n = \frac{2}{T} \int_{-T/2}^{T/2} f(x) \cos \frac{2\pi n}{T} x\, dx$$

$$b_n = \frac{2}{T} \int_{-T/2}^{T/2} f(x) \operatorname{sen} \frac{2\pi n}{T} x\, dx$$

$$a_0 = \frac{2}{T} \int_{-T/2}^{T/2} f(x)\, dx$$